The final mesh interval of the sampled-data solvers in `voles`

Implementation note (voles 1.0)

14 November 2026

The array-input solvers (`solve_vie1/2/vide` with sampled `kernel`, `g`, `a`) use a uniform mesh of Q samples per interval. The record need not end on a mesh point: when $N-1$ is not a multiple of Q, the last regular interval and the r leftover samples are merged into one longer *final interval*, with its own collocation nodes snapped to the sample grid and its own collocation row, solved after the regular intervals. No data are discarded. This note describes that construction as implemented in `_product.mesh_split`, `_product.snap_nodes`, `_product.solve_tail`, `_sampled._with_tail` and `_product._finish`.

1 Notation

Symbol	Meaning	Code
δ	sample spacing	`dt`, `delta`
N	number of samples; $t_j = j\delta$, $j = 0, \dots, N-1$; $T = (N-1)\delta$	`N`
$K_j,\ g_j,\ a_j$	data samples $K(t_j)$, $g(t_j)$, $a(t_j)$ (scalars or $d \times d$ / d blocks)	`kernel`, `g`, `a`
q	number of divisions of the reference interval	`divs`
$k_1 < \cdots < k_m$	the chosen node indices, $k_i \in \{0, \dots, q\}$; m nodes	`choices`
$c_i = k_i/q$	nominal collocation nodes in $[0,1]$	`nodes`
Q	samples per regular interval (a multiple of q)	`samples_per_interval`
$H = Q\delta$	width of a regular interval	
$M_{\mathrm{reg}},\ r$	$N-1 = M_{\mathrm{reg}}Q + r,\ \ 0 \le r < Q$	
M	number of regular intervals: M_{reg} if $r = 0$, else $M_{\mathrm{reg}} - 1$	`M`
$t_l = lH$	mesh points; regular intervals $I_l = [t_l, t_{l+1}]$, $l = 0, \dots, M-1$	
$Q' = Q + r$	samples in the final interval ($Q < Q' < 2Q$)	`Qp`
$h = Q'\delta$	width of the final interval $I_M = [t_M, T]$, $\ H < h < 2H$	
κ_i	sample offset of the i-th node in I_M (the snapped node)	`kap`
$c'_i = \kappa_i/Q'$	snapped local nodes; collocation points $\tau_i = t_M + \kappa_i\delta$	`node_times[-1]`
v	local variable on I_M: $s = t_M + vh$, $v \in [0,1]$	
$L_k(v)$	Lagrange basis on $\{c'_1, \dots, c'_m\}$ (on $\{0, c'_1, \dots, c'_m\}$ for the continuous VIE-1)	`basis_p`
u_l	collocation solution on I_l (local monomial coefficients)	`unit_reg`, `unit_t`
U_k	unknowns of the final row: $U_k = u_M(\tau_k)$	`U`
p	kernel interpolation degree: `kernel_interp_degree` (product), m (collocation)	`p`
K_h	piecewise degree-p interpolant of the samples K_j (below)	`coef`
η_i	history term at τ_i (contribution of $I_0, \dots, I_{M-1}$)	`hist`
A_{ik}	local block: weight of U_k in the integral over $[t_M, \tau_i]$	`Ab`
y_{prev}	$u_{M-1}(t_M^-)$, or y_0 if $M = 0$ (continuous VIE-1, VIDE)	`y_prev`

2 Splitting the record

`mesh_split(N, Q)` returns (M, Q') from

$$N - 1 = M_{\mathrm{reg}}\, Q + r, \qquad (M, Q') = \begin{cases} (M_{\mathrm{reg}},\ 0) & r = 0 \quad \text{(no final interval)}, \\ (M_{\mathrm{reg}} - 1,\ Q + r) & r > 0. \end{cases}$$

The final interval is therefore the last *full* interval stretched over the leftover samples, not a short interval of r samples: its width stays within $(H, 2H)$. (A short interval of $r \ll Q$ samples would make the first-kind

local system nearly singular, since $A = O(h)$.) If $M_{\text{reg}} = 1$ and $r > 0$ the whole record is a single stretched interval ($M = 0$). The only constraint is $N \geq Q + 1$. The regular intervals are solved exactly as before (D or Numba driver for `quadrature="collocation"`, the lag-block driver for `"product"`); the final row is assembled and solved afterwards in NumPy.

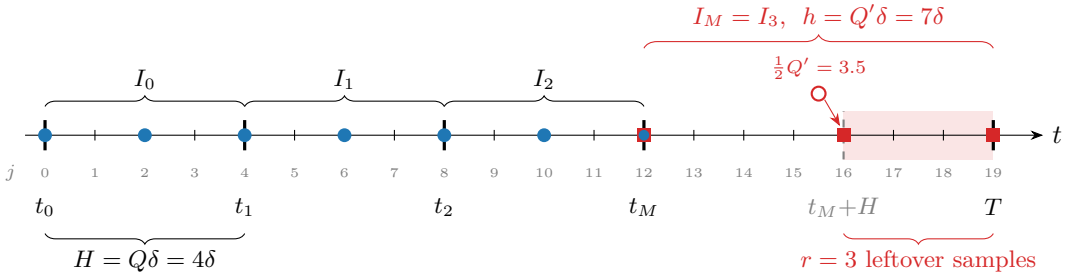


Figure 1: Splitting for $(q, \{k_i\}) = (2, \{0, 1, 2\})$ with `quadrature="collocation"` ($Q = q^2 = 4$) and $N = 20$: $19 = 4 \cdot 4 + 3$, so $M = 3$ regular intervals and a final interval of $Q' = 7$ samples. Blue circles: collocation points of the regular intervals (sample offsets $k_i Q/q = 0, 2, 4$). Red squares: snapped nodes of the final interval, $\kappa = (0, 4, 7)$; the nominal midpoint 3.5 (hollow) is a tie and rounds up to 4. The dashed line is where the last regular mesh point would have been.

3 Snapping the nodes

On I_M the nominal node c_i would lie $k_i Q'/q$ samples from t_M, generally not a sample. `snap_nodes` moves it to the nearest sample, ties rounding up:

$$\kappa_i = \left\lfloor \frac{k_i\, Q'}{q} + \frac{1}{2} \right\rfloor, \qquad c'_i = \frac{\kappa_i}{Q'}, \qquad \tau_i = t_M + \kappa_i\, \delta.$$

Properties:

- $k_i = 0 \mapsto \kappa_i = 0$ and $k_i = q \mapsto \kappa_i = Q'$: the end nodes stay at the interval ends.
- $|c'_i - c_i| \leq 1/(2Q')$, i.e. each collocation point moves by at most $\delta/2$.
- The κ_i are distinct: since $Q' > Q \geq q$, consecutive scaled nodes differ by more than one sample and rounding is monotone.
- Exact in floating point: a tie is an integer plus $\frac{1}{2}$ (exactly representable); a non-tie is at least $1/(2q)$ from a half-integer.
- Ties occur only for even q, when $k_i Q'/q$ has denominator 2. All ties move right, so a node set symmetric under $c \mapsto 1 - c$ generally loses that symmetry.
- g and a are only needed at samples ($g_i = g_{MQ+\kappa_i}$), so they are never interpolated; K enters the local blocks through its interpolant K_h.

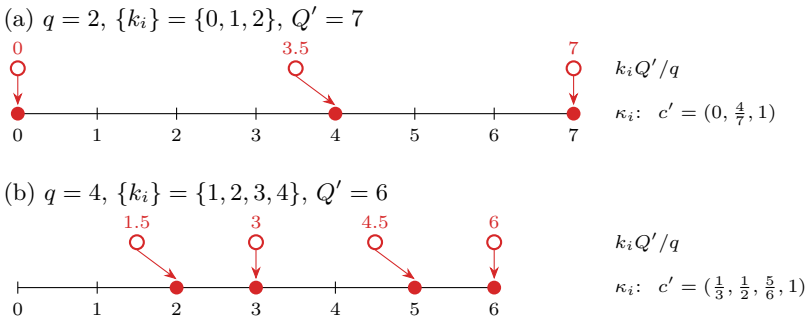


Figure 2: Snapping on the final interval. Hollow: nominal positions $k_i Q'/q$ in samples; filled: κ_i. In (b) both ties (1.5, 4.5) move right; the nominal pair $\{\frac{1}{4}, \frac{3}{4}\}$, symmetric about $\frac{1}{2}$, becomes $\{\frac{1}{3}, \frac{5}{6}\}$.

Setting	Q	snapped c' by r
$(2, \{0,1,2\})$, product (VIE-2/VIDE default)	2	$r=1$: $(0, \frac{2}{3}, 1)$ (tie)
$(2, \{0,1,2\})$, collocation	4	$r=1$: $(0, \frac{3}{5}, 1)$; $r=2$: $(0, \frac{1}{2}, 1)$; $r=3$: $(0, \frac{4}{7}, 1)$
$(3, \{1,2,3\})$, product (VIE-1 default)	9	no ties; e.g. $r=1$: $(\frac{3}{10}, \frac{7}{10}, 1)$, $r=3$: $(\frac{1}{3}, \frac{2}{3}, 1)$

4 The collocation row of the final interval

We write the VIE-2, $y(t) = g(t) + \int_0^t K(t-s)\,y(s)\,ds$; the other kinds follow at the end. On I_M the approximation is $u_M(t_M + vh) = \sum_{k=1}^{m} U_k L_k(v)$, and the collocation equations at $\tau_1, \dots, \tau_m$ split the integral at t_M:

$$U_i \;=\; g(\tau_i) \;+\; \underbrace{\sum_{l=0}^{M-1} \int_{I_l} K(\tau_i - s)\,u_l(s)\,ds}_{\eta_i \text{ (history)}} \;+\; \underbrace{\sum_{k=1}^{m} U_k \int_{t_M}^{\tau_i} K_h(\tau_i - s)\,L_k\Big(\frac{s - t_M}{h}\Big)\,ds}_{\sum_k A_{ik} U_k \text{ (local)}}, \qquad i = 1, \dots, m,$$

i.e. the $m \times m$ (or $md \times md$) system $(I - A)\,U = g(\tau) + \eta$. The u_l, $l < M$, are known.

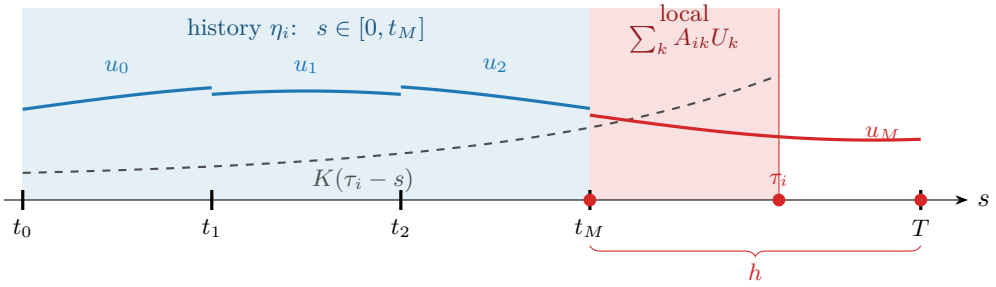


Figure 3: The two parts of the integral at a collocation point τ_i of the final interval (schematic, $M = 3$). Over the regular intervals (blue) the known pieces u_l, which may jump at the mesh points, are integrated against the kernel to give η_i; over $[t_M, \tau_i]$ (red) the unknown u_M enters through the local blocks A_{ik}.

Kernel interpolant. On each cell $[t_j, t_{j+1}]$, K_h is the degree-p polynomial through the $p+1$ samples nearest the cell (one-sided at the ends of the array) (`interp_cell_coefs`). With $\Lambda_{k,c,\rho} = \int_{\text{cell } c} L_k(v)\,\xi^\rho$ the moment tensor of the snapped basis on the Q' cells of I_M ($\xi \in [0,1]$ the cell variable), the local blocks are exact integrals of piecewise polynomials:

$$A_{ik} = \delta \sum_{c=0}^{\kappa_i - 1} \sum_{\rho=0}^{p} K_{h,\rho}^{(\kappa_i - 1 - c)}\,\Lambda_{k,c,\rho}, \qquad A_{ik} = 0 \;\text{ if } \kappa_i = 0,$$

where $K_{h,\rho}^{(j)}$ is the ρ-th monomial coefficient of K_h on cell j (`_tail_local_blocks`). The local blocks use product integration on *both* quadrature paths.

History term. This is where the two quadratures differ.

- `quadrature="product"`: $\eta_i = \sum_{l<M} \int_{I_l} K_h(\tau_i - s)\,u_l(s)\,ds$, integrated exactly, with K_h the interpolant of the *full* sample array (`_tail_history`).
- `quadrature="collocation"`: the interpolatory rule the regular rows use,

$$\eta_i = H \sum_{l=0}^{M-1} \sum_{k} b_k\,K(\tau_i - t_l - \hat{c}_k H)\,u_l(t_l + \hat{c}_k H), \qquad b_k = \int_0^1 \ell_k(v)\,dv,$$

  with ℓ_k the Lagrange basis on the regular nodes $\hat{c} = (c_1, \dots, c_m)$ (prefixed by 0 for the continuous VIE-1) (`_tail_history_node_rule`). Every kernel argument $\big((M-l)Q + \kappa_i - \hat{k}Q/q\big)\delta$ is a sample. Using the same rule as the regular rows matters for the first kind: a quadrature mismatch between rows is amplified by $\sim 1/H$ and was measured to cost one order.

The other equation kinds. With $g_i = g(\tau_i)$:

- VIE-1 (discontinuous): $A\,U = g - \eta$.
- VIE-1 (continuous): basis $L_0, \dots, L_m$ on $\{0, c'_1, \dots, c'_m\}$ with $U_0 = y_{\text{prev}}$ fixed: $\sum_{k\ge 1} A_{ik} U_k = g_i - \eta_i - A_{i0}\, y_{\text{prev}}$.
- VIDE, $y' = a\,y + g + \int_0^t K(t-s)\,y(s)\,ds$: $u_M(t_M + vh) = y_{\text{prev}} + h\sum_k Y_k\,\beta_k(v)$, $\beta_k(v) = \int_0^v L_k$, $Y_k = u'_M(\tau_k)$, and

$$Y_i - \sum_k \big[A_{ik} + h\,a(\tau_i)\,\beta_k(c'_i)\big] Y_k = g_i + \eta_i + \big(A_{i0} + a(\tau_i)\big)\,y_{\text{prev}},$$

  where here A_{ik} integrates K_h against $h\beta_k$ and $A_{i0} = \int_{t_M}^{\tau_i} K_h(\tau_i - s)\,ds$.

For systems ($d > 0$) every entry is a $d \times d$ block, and the system is $md \times md$.

5 Assembling the output

- Samples $j \le MQ - 1$: from the regular intervals. Samples $j = MQ + 1, \dots, N - 1$: $y_j = u_M(t_j)$.
- The shared mesh point $j = MQ$: $y_{MQ} = \frac{1}{2}\big(u_{M-1}(t_M^-) + u_M(t_M^+)\big)$ (the same convention as at every interior breakpoint of a discontinuous solution).
- The `VolterraSolution` has $M + 1$ pieces with `mesh` $= (0, H, \dots, MH, T)$; `node_times[-1]` holds the snapped τ_i, while `nodes` keeps the nominal c_i.
- With `quadrature="collocation"` the regular part is computed from `kernel[:MQ+1]` and is bit-identical to a solve on the truncated record. With `"product"`, K_h is built from the full array, so cells near t_M get centred rather than one-sided stencils; the regular part then differs from a truncated solve at the level of the kernel-interpolation error (about 2×10^{-11} in the example of Fig. 4).

6 Accuracy

The final interval is one extra interval of width below $2H$ with a bounded node set, so it adds one local error of the method's order and leaves the global order unchanged (`tests/test_final_interval.py`). For the default VIE-2 ($m = 3$, local polynomials of degree 2, $Q = q$, so every regular sample is a node and converges at the superconvergent order 4):

	tail offset 1	tail offset 2	final point T
code (ties up), $c' = (0, \frac{2}{3}, 1)$	3.00 (not a node)	3.99 (node)	3.99
ties down (patched), $c' = (0, \frac{1}{3}, 1)$	3.99 (node)	2.99 (not a node)	3.99

Collocation points converge at order $m + 1 = 4$; the one sample of I_M that is not a node converges at the interpolation order 3. The final point keeps order 4 for *any* snapped node set because $c' = 1$ is a node, not because the Simpson superconvergence survives snapping. The rounding direction only chooses which sample is off-node; the error constants differ by about 10%. When $Q > q$ (e.g. `quadrature="collocation"`), regular intervals already contain off-node samples, so the final interval is no worse than the rest.

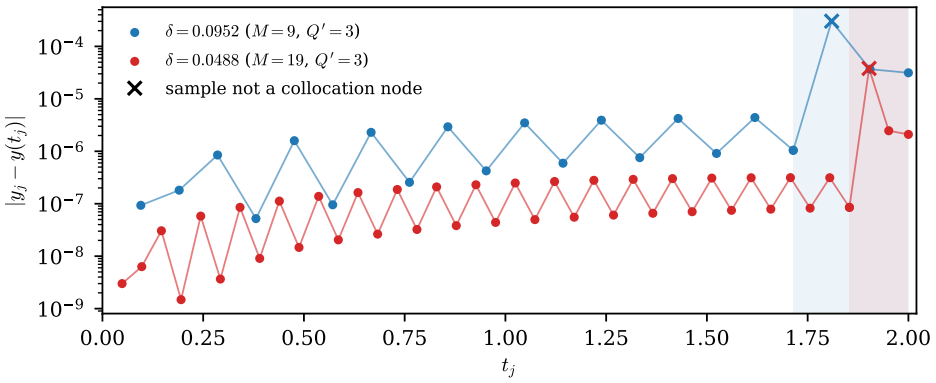


Figure 4: Error by sample for $y(t) = g(t) + \int_0^t e^{-(t-s)} y(s)\, ds$ with exact solution $y = \cos t$, default settings (product, $(2, \{0, 1, 2\})$, $Q = 2$), $T = 2$ and N even, so $Q' = 3$ and $c' = (0, \frac{2}{3}, 1)$. Shaded: the final interval. Crosses: the tail sample at offset 1, which is not a collocation node; it is the largest error and converges one order lower (between the two resolutions its error falls by about 2^3, the others by close to 2^4). Generated by `error_figure.py`.